

Credit Risk Analytics

Novabank

Default drivers, risk segmentation and the profile of the borrowers most likely to default — 32,574 approved loans and \$312M of gross exposure across the UK, Canada and the US.

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Portfolio risk position

Portfolio at a Glance

32,574

Loans in scope

\$312M

Gross exposure

21.82%

Portfolio default rate

\$34.69M

Expected loss

One loan in five has defaulted; expected loss runs at 11.11% of exposure, and \$77.09M of exposure sits on defaulted loans.

Scope and Data

Population

32,574 approved loans across three markets, \$312M of gross exposure, \$9,588 average loan, none above \$35,000.

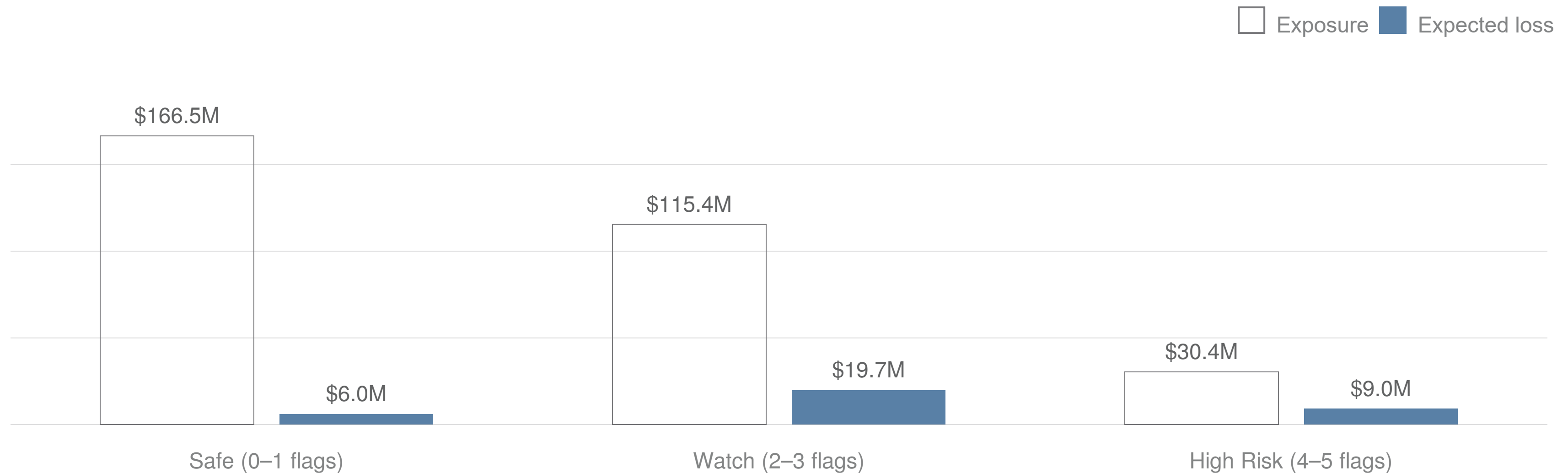
Risk flags

Each loan carries 0–5 flags - high LTI, high DTI, low income, short tenure, prior default - grouped into Safe, Watch and High Risk.

Measures

Default rate is realised defaults over loans. Expected loss rate is modelled loss over exposure. Pricing gap is interest rate less expected loss rate.

Expected Loss Concentration



The Watch segment holds 36.9% of exposure but produces 56.8% of expected loss.

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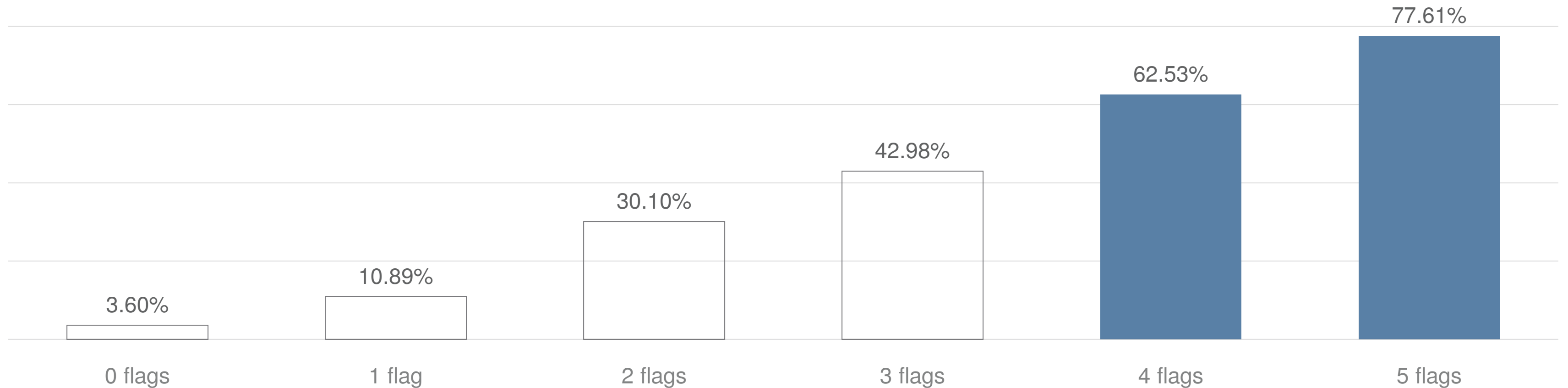
Risk segmentation

The Three-Segment Framework

SEGMENT	CUSTOMERS	SHARE	DEFAULT RATE	EXPOSURE	% OF EXPOSURE
Safe — 0–1 flags	18,405	56.5%	7.36%	\$166.5M	53.3%
Watch — 2–3 flags	11,410	35.0%	34.59%	\$115.4M	36.9%
High Risk — 4–5 flags	2,759	8.5%	65.46%	\$30.4M	9.7%
Portfolio	32,574	100%	21.82%	\$312.3M	100%

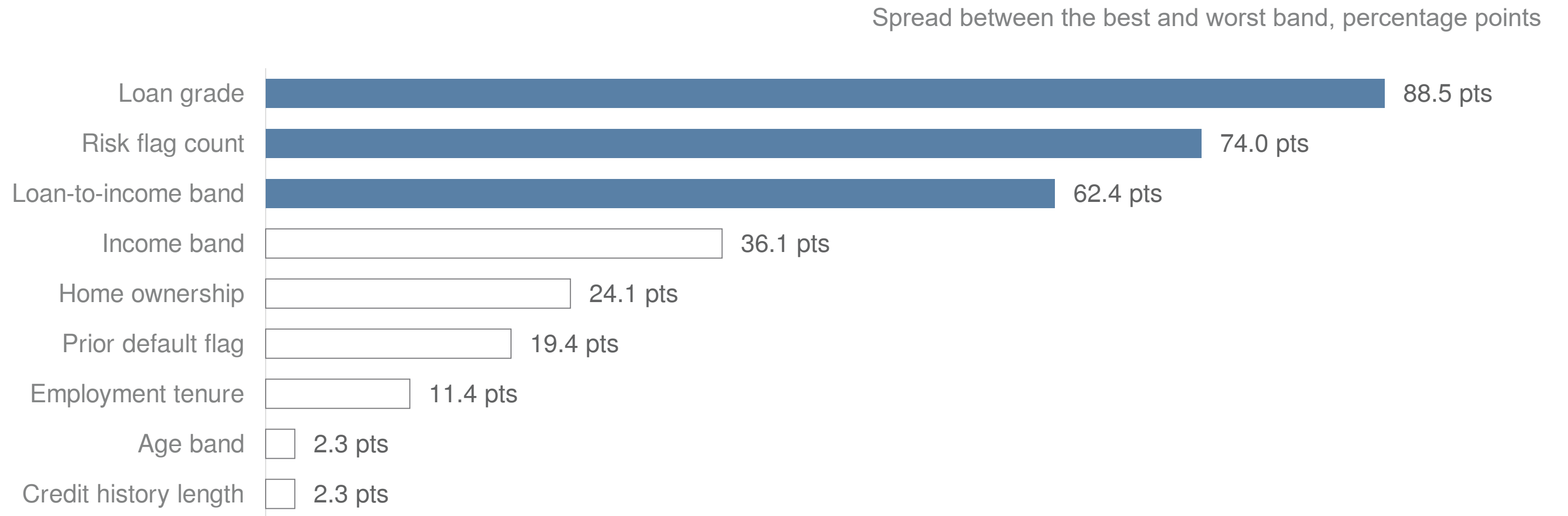
The 8.5% of customers carrying four or five flags default at 65.46% - nine times the Safe segment's rate, and the flag count alone sorts the book before any grade or price is applied.

Default Rate by Risk Flag Count



3.60% at zero flags against 77.61% at five — the sharpest single ranking available.

Predictive Power by Variable



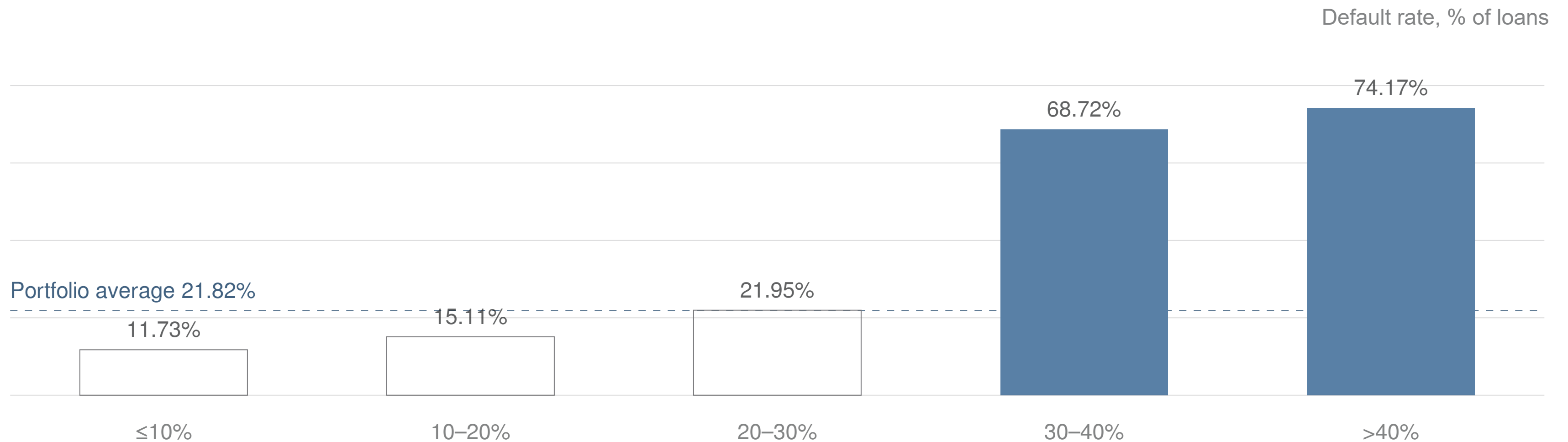
Three variables do the work — grade, flag count and loan-to-income. Age and credit history separate nothing.

Loan-to-Income Above 40%

70.31%

Default rate on the 3,833 loans where the loan amount exceeds 40% of annual income, against a 21.82% portfolio average.
Average DTI across the book is 34.52%.

Default Rate by Loan-to-Income Band



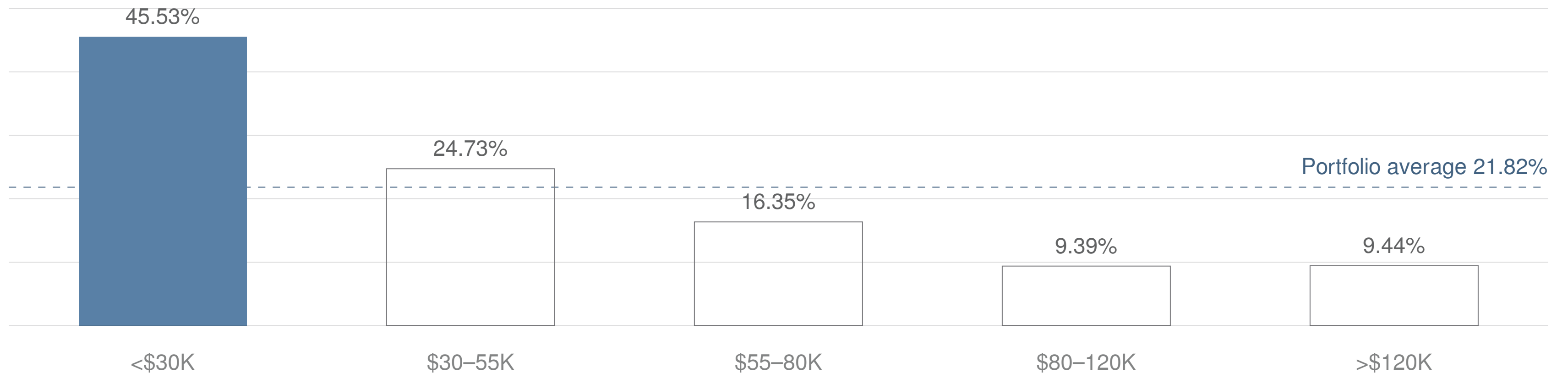
Rates hold near the portfolio average up to a 30% ratio, then triple above it.

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Borrower profile

Default Rate by Income Band

Default rate, % of loans



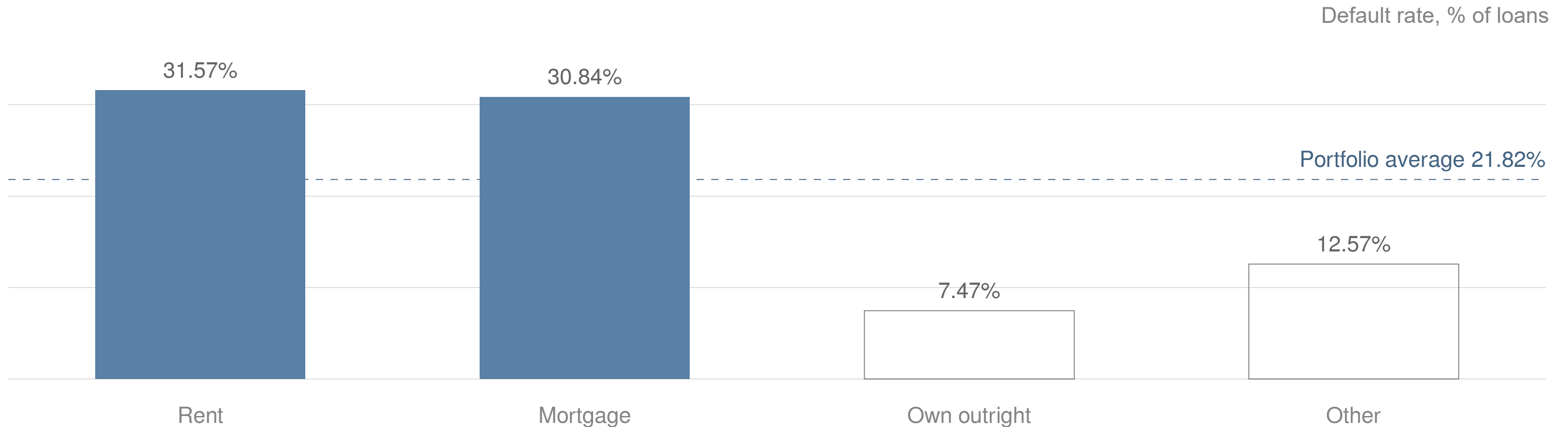
Borrowers under \$30K default at 45.53%, five times the rate above \$80K.

Income and Home Ownership

INCOME BAND	MORTGAGE	OWN OUTRIGHT	RENT	OTHER
<\$30K	27.95%	25.16%	53.01%	66.67%
\$30–55K	15.34%	1.22%	32.96%	42.42%
\$55–80K	12.89%	2.66%	21.85%	23.53%
\$80–120K	7.87%	0.75%	13.59%	11.76%
>\$120K	8.65%	2.40%	13.70%	9.09%

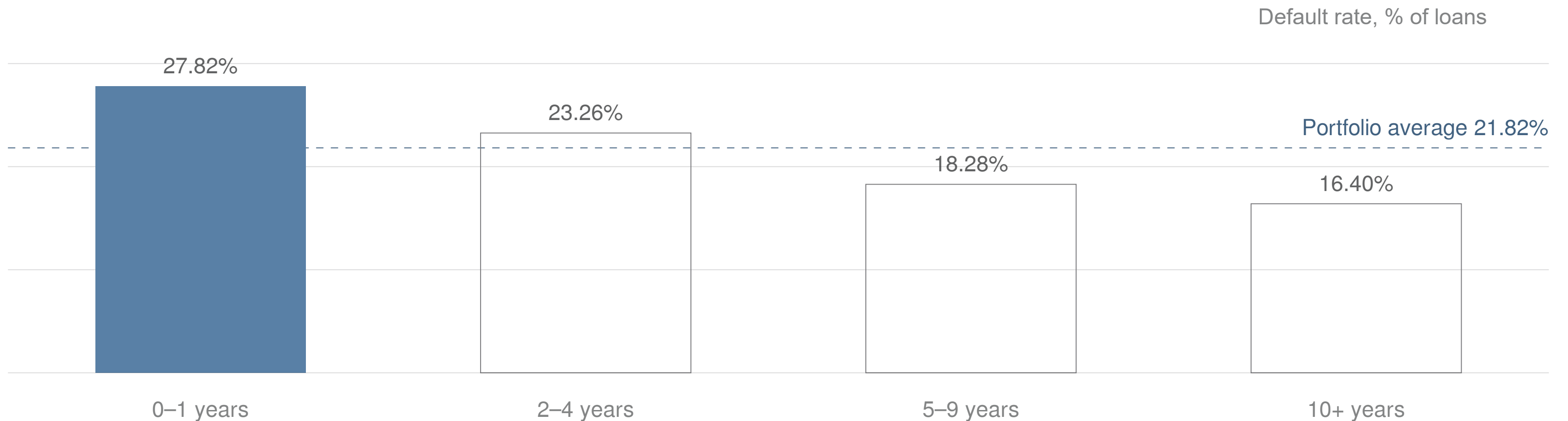
Renting compounds low income: a renter under \$30K defaults at 53.01% against 27.95% for a mortgage holder in the same band. Borrowers who own outright stay below 3% once income passes \$30K.

Default Rate by Home Ownership



Mortgage holders read as risky in total, yet beat renters inside every income band — a mix effect, not a housing effect.

Default Rate by Employment Tenure



Default rate falls with tenure, from 27.82% in the first year to 16.40% after ten.

Variables With Little Signal

VARIABLE	RANGE ACROSS BANDS	SPREAD	READ
Age — 20–25 to 46+	23.02% → 20.72%	2.3 pts	No underwriting value
Credit history length — 0–3 to 10+ years	22.93% → 21.01%	1.9 pts	No underwriting value
Past delinquencies — 0 to 5	21.77% → 21.19%	flat, thin cells	Not predictive on its own
Prior default flag	18.40% → 37.80%	19.4 pts	Keep — the one history variable that separates

Demographics and bureau history carry almost no signal in this book. Capacity variables — income, loan-to-income, DTI and tenure — carry nearly all of it, alongside the prior-default flag.

The High-Risk Borrower Profile

Capacity

Income under \$30K, loan-to-income above 40%, DTI above the 34.52% portfolio average.



Tenure

Under two years of employment history, most often inside the first year.



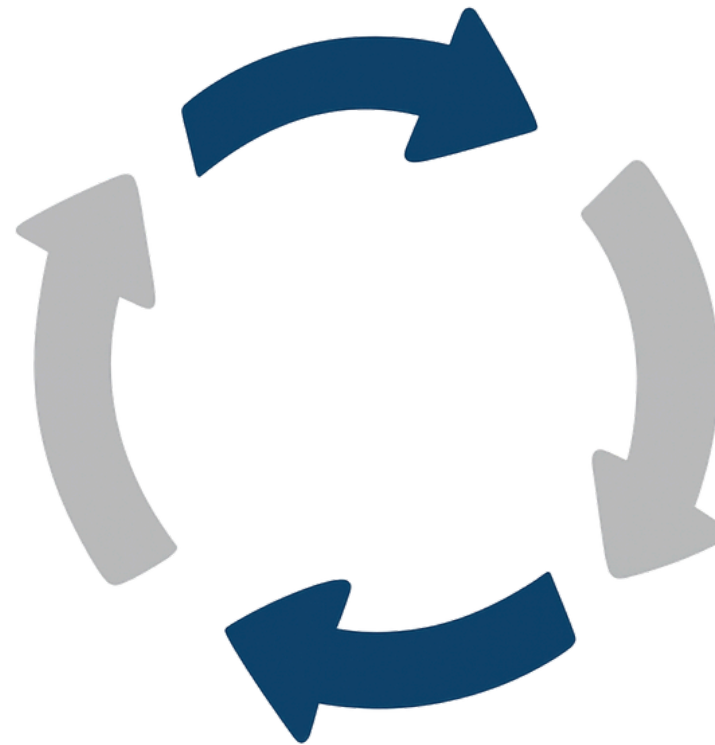
Housing

Renting or in the “other” housing category is no mortgage and no owned property.



Grade and history

Graded D or below, frequently carrying a prior default flag: 2,759 customers, \$30.4M of exposure.

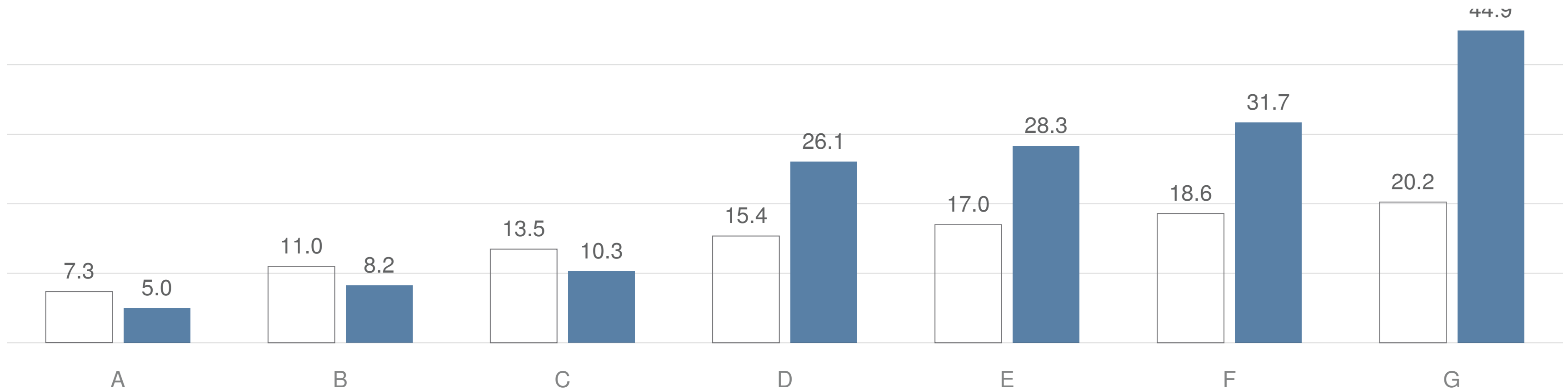


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Pricing and recommendations

Pricing vs Expected Loss

 Interest rate
  Expected loss



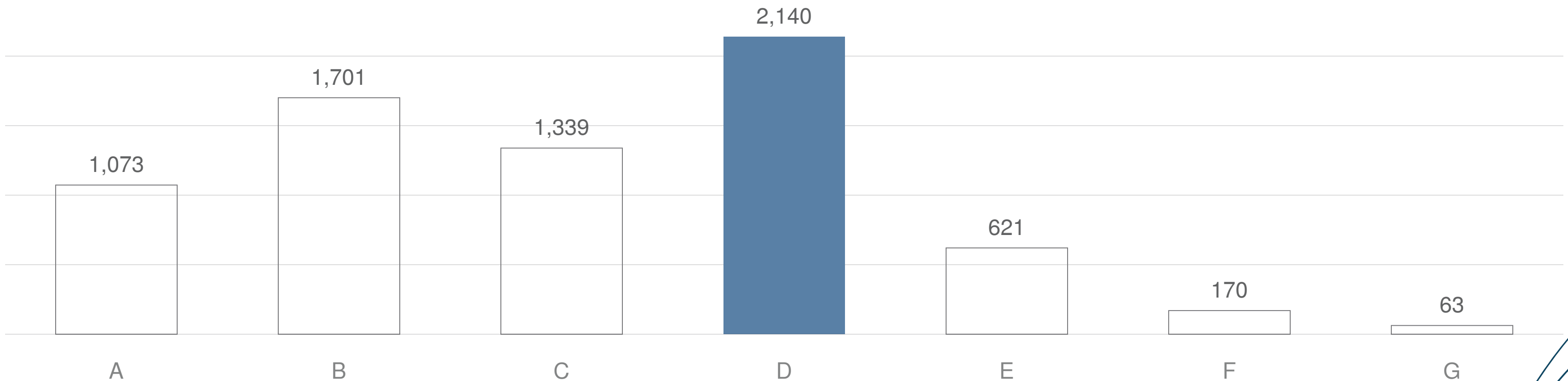
From grade D the expected loss rate runs 10 to 25 points above the rate charged.

Grade Economics

GRADE	CUSTOMERS	DEFAULT RATE	AVG INTEREST RATE	EXPECTED LOSS RATE	PRICING GAP
A	10,776	9.96%	7.34%	4.99%	+2.35
B	10,448	16.28%	10.99%	8.24%	+2.75
C	6,456	20.74%	13.47%	10.27%	+3.19
D	3,625	59.03%	15.36%	26.06%	-10.71
E	964	64.42%	16.99%	28.29%	-11.29
F	241	70.54%	18.60%	31.68%	-13.08
G	64	98.44%	20.24%	44.93%	-24.69
Portfolio	32,574	21.82%	11.01%	11.11%	-0.09

Where the Defaults Come From

Defaulted loans, derived from grade counts and rates

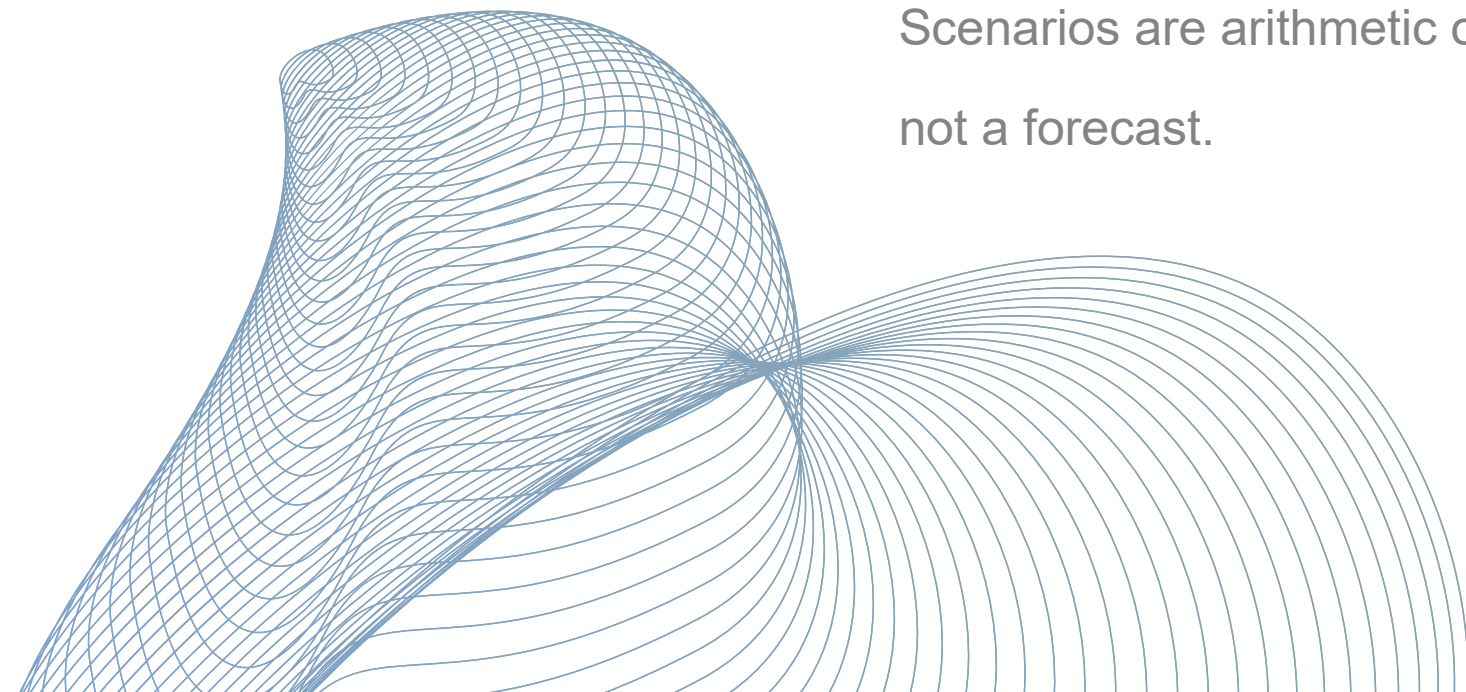


Grade D is 11.1% of customers but 30.1% of all defaults — the largest single block of loss in the book.

Loss Reduction Scenarios

ACTION	POPULATION	EFFECT ON LOSS	TRADE-OFF
Exit the High Risk segment	2,759 · \$30.4M	–\$9.0M expected loss (26%)	8.5% of customers
Cap loan-to-income at 30%	3,833 loans	≈2,695 defaults avoided	Declines under \$30K income
Re-price grades D–G	4,894 · 15.0%	Closes the –0.09% gap	+10 to 25 pts of margin
Monthly Watch-band review	11,410 · \$115.4M	Addresses \$19.7M (56.8%)	Monitoring cost only

Scenarios are arithmetic on the dashboard's own figures and assume observed rates hold; they are sizing, not a forecast.



Underwriting Recommendations



CAP LOAN-TO-INCOME AT 30%

Decline or restructure above: 3,833 loans breach 40% today and default at 70.31%. Hardest where income is under \$30K and the borrower rents.



RE-PRICE OR EXIT D-G

15.0% of customers, and all of the negative pricing gap. Each grade needs 10 to 25 points of additional margin, a lower limit, or no offer.



REVIEW THE WATCH BAND

11,410 customers at 34.59% hold \$115.4M and generate 56.8% of expected loss. A change in flag count should trigger monthly review before delinquency appears.

DISCUSSION

Dataset: NovaBank loan book, 32,574 approved loans across the UK, Canada and the US, reported through the Credit Risk dashboard.

Source: Xóm data